ODE-FFN: Progress Report 2

Independent Study — Deep Learning

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1 Project and progress

This project asks whether continuous-flow structure can make a useful FFN inside JiT. Starting from two Taylor maps around SiLU, we explored clocks, generator P, polynomial coupling, nonlinear response, full-field readouts and learned directions. Several changes help specific tasks; ordinary controls limit what can be credited to the ODE construction.

Tests use a width-64, three-block, time-conditioned vector MLP, without JiT attention or patches. Independent noise–data pairs train velocity from $z_t = tX + (1-t)e$ on analytic mixtures. Oracle MSE measures fitting; test SW2 measures generation. Earlier studies use 5k updates; recent screens use 20k, batch 512 and Heun32. Comparisons retain their own protocols.

2 Local dynamics: clocks, order and optimization

The original scalar map at $K = 2$ is

$$T(h) = h + s(h)Ph + \tfrac{1}{2}s(h)^2P^2h, \qquad s(h) = 0.2 + 0.8\,\sigma(g(h)/10). \tag{1}$$

Clocks and order. The input clock differs from FM time and contributes to the full Jacobian. On Ring8, input gates reduce samples beyond 0.6 from every mode center: 19.9% to 8.3%, versus SwiGLU's 3.4% (parameter counts differ). Freeing fixed-clock effective matrices improves fitting. Exact frozen solves, state feedback, clock reuse, biases and augmented states yield no general quality–cost gain. Scalar K1 improves every matched K1/K2 field comparison; raw K1 at 80k still loses sampling or wider controls. Useful adaptation does not establish a benefit of tied higher powers.

Optimization. Muon approximately orthogonalizes FFN momentum updates; other parameters retain AdamW. We measured both updates and learned P.

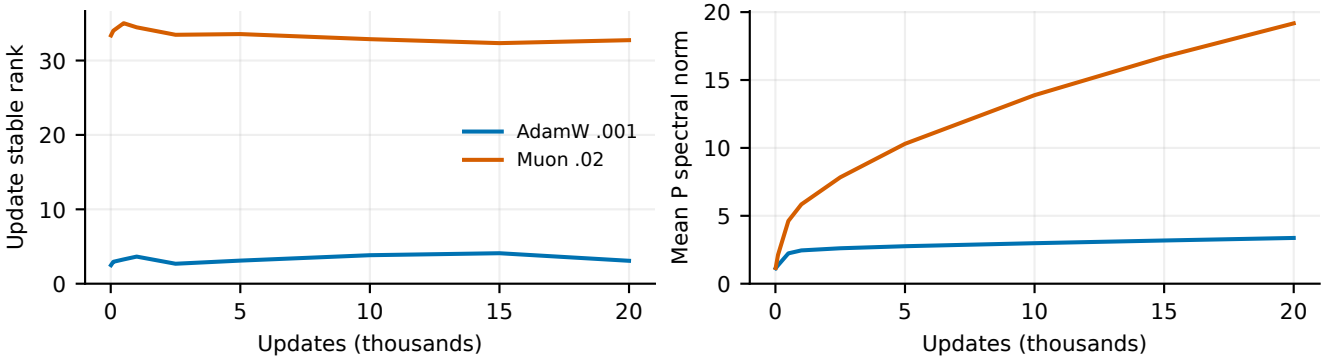


Figure 1: Product16 scalar K2, six-matrix means. Matrix rates: AdamW .001, Muon .02. Update stable rank is $\|\Delta P\|_F^2/\|\Delta P\|_2^2$. Broader updates coexist with growing P.

At 20k, AdamW/Muon update stable ranks are 3.08/32.74 and mean P norms 3.37/19.17. Field MSE improves from 0.03343 to 0.02385; SwiGLU192 reaches 0.01567. Norm-matched momentum does not reproduce all gains. Decay and training-time flattening harm fitting; flattening also worsens SW2.

3 Changing the complete field and its directions

Full-gradient readout. We tested whether structure in the complete predicted field matters more than a constraint on hidden P. For a vector backbone G, predict

$$v(x,t) = \nabla_x[\tfrac{1}{2}x^\top G(x,t)] = \tfrac{1}{2}[G(x,t) + J_G(x,t)^\top x]. \tag{2}$$

This matches the gradient-field structure of the analytic mixture velocities. At 5k, original-ODE Product16 MSE falls from 0.325527 to 0.037320. However, ordinary gradient SwiGLU reaches 0.033826 at nearly equal dense inference work. Derivative-path and local-adjoint controls show that localizing this operation does not recover its full-network gain. Gaussian sampling can worsen despite better field fitting. The result identifies a useful computation, with extra work and no general ODE-specific advantage.

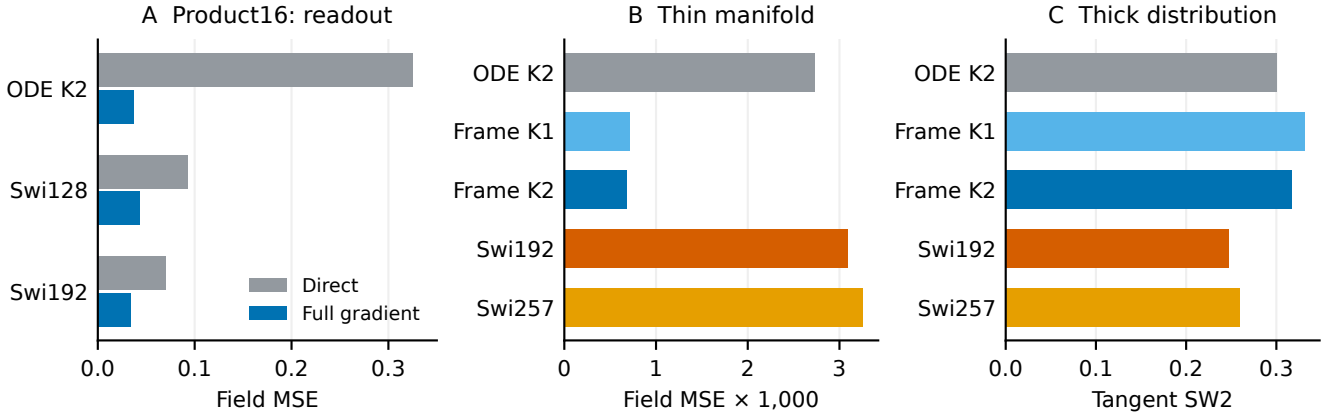


Figure 2: Lower is better. A: full-gradient readout helps both ODE and ordinary backbones. B: conditional frames improve thin-target field fitting. C: their sampling advantage fails on the thick target. Each panel has its own metric and scale; scores are not pooled.

Input-dependent directions. To change the directions in which P acts without changing its pointwise spectra, we tested $Q(h)T_K(s(h)P)Q(h)^\top h$ with an orthogonal learned frame. On a 2D mode plane embedded in 128D, frame K2 reaches MSE 0.000685 and K1 reaches 0.000713, versus SwiGLU192's 0.003095. K1 already has better tangent SW2 than K2. Adding variance outside the plane reverses the sampling result: frame K2 gives 0.3173 versus SwiGLU192's 0.2471. Quadratic-action, independent-second-coefficient and gradient-route follow-ups do not establish a transferable K2 benefit. The shared output-head bottleneck also limits interpretation of thick-task errors.

Nonlinear internal flow. We tested whether linear dynamics were too restrictive by splitting $dy/dr = Py - y^{\odot 3}$ into linear steps and the exact response $\phi_s(u) = u/\sqrt{1+2su^2}$. Product16 field fitting improves, but freeing the two matrices around that response works better:

5k Product16	Original K2	Tied cubic	Free ISRU	SwiGLU160
Field MSE	0.325527	0.092938	0.037159	0.078178

Table 1: Same earlier protocol. Free ISRU is an ordinary independent-matrix control, with a different parameterization and preserved special initializer.

Its advantage reverses on dependent-sign and half-mixture laws. Noisy-label, affine-head, frozen-feature and local-teacher studies likewise fail to explain or eliminate the whole ordinary-model gap. These are finite-budget diagnoses, not approximation lower bounds. Exact ordinary-flow realizations further show that an ODE interpretation alone does not distinguish a useful FFN.

4 Input-dependent features and current evidence

The diagonal construction uses $D(h) = \mathrm{diag}(d(h))$ and $A(h) = PD(h)$. At K1,

$$F_1(h) = h + P[d(h) \odot h], \qquad J_{F_1} = I + P[\mathrm{diag}(d) + \mathrm{diag}(h)J_d]. \tag{3}$$

Thus conditional matrix spectra omit input feedback. With a free affine controller, $P' = PC^{-1}, D' = CD$ can preserve PD while changing P's spectrum. For $q = D(h)h$, increment energy is $\mathrm{tr}(PMP^\top)$ with $M = \mathbb{E}[qq^\top]$; feature statistics matter alongside matrix shape.

Question	Designs, purpose and observations
Controller or output range?	MLP/rank-32 linear controllers, sigmoid/affine output and frozen swaps separate capacity from range. Weak-parent gains do not beat near-parameter SwiGLU.
Which feature function?	Constant, h, sigmoid, $h+1$ and $\lvert h\rvert$ test linear/quadratic and sign effects. Raw K1 has a small Ring8 field lead with worse sampling; absolute input worsens matched fits.
Does smaller growth help?	Radial versus raw and fixed scaling preserves feature direction while changing amplitude. Responses shrink without a qualifying gain.
Can a bounded odd response help?	$d_i = (1 + 1.2h_i^2)^{-1/2}$ gives $q_i = h_i/\sqrt{1 + 1.2h_i^2}$. Compare two retained solves around SiLU with constant D, ordinary free ISRU and SwiGLU.

The last design follows the earlier ISRU lead, not an assumption that bounded features guarantee stable generation. Constant D shares every initial tensor; free ISRU changes the FFN initial function. Representative 20k endpoints are:

	Ring8		Product16	
FFN	MSE	SW2	MSE	SW2
ISRU-D	**0.00977**	**0.14591**	0.01606	0.08574
Free ISRU	0.01000	0.17339	**0.01536**	0.08432
SwiGLU64	0.01128	0.15038	0.01878	0.08106
SwiGLU192	0.01123	0.16580	0.01567	**0.07921**

Table 2: ISRU arms: AdamW .0003. Earlier validation-selected SwiGLU: Muon .02 on Ring8; AdamW .0003 on Product16, except Swi64 at .001. ISRU-D/free have 58,512 Product16 parameters versus Swi64's 58,572. The full grid retains constant D, Swi128 and both optimizer settings; lower is better.

ISRU-D improves Ring8 fitting and sampling over all three SwiGLU widths, but its field lead over free ISRU is only 2.25%, below the registered 5% continuation margin. Product16 favors ordinary controls. Muon worsens both ISRU fits at the tested rate, unlike its benefit for the original scalar model. These results support a response/optimizer interaction, not a causal spectral explanation. Image transfer remains unvalidated; an adequately trained ordinary baseline below the project's FID-50 gate is required. The full attempt ledger and evidence index preserve every earlier study and comparison limit.

5 Hypotheses for the next two weeks

1. **Feature response and optimization coordinates explain more than raw P regularity.** At saved states, compare native and norm-matched updates on the same minibatch. Measure changes in Pq and full-field loss alongside spectra before proposing another constraint or polynomial coupling.
2. **Task-specific gains depend on where errors occur during transport.** Compare saved ISRU-D, free-ISRU and ordinary-control checkpoints against the analytic field at common times and generated-path states. Fix checkpoints and solver resolution to separate error location from average MSE, as the readout and thin/thick studies already distinguish fitting from generation.